

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

ANSWERS

Question 1: Compare Star Topology and Bus Topology

Introduction:

Star and Bus are two common LAN topologies. Star topology connects all devices to a central switch or hub, while Bus topology connects all devices through a single backbone cable. They differ in cost, reliability, performance, and maintenance.

Comparison Table:

Parameter	Star Topology	Bus Topology
Structure	All devices connect to a central switch	All devices share one backbone cable
Cost	High (more cable and switch required)	Low (single backbone cable)
Installation	Easy but requires more cabling	Simple and inexpensive
Performance	High, fewer collisions	Performance decreases as devices increase
Reliability	High; failure of one cable affects only one device	Low; backbone cable failure stops the entire network
Maintenance	Easy to troubleshoot	Difficult to locate faults
Scalability	Easy to expand	Difficult to expand
Example	Offices, Schools	Small laboratories

Conclusion:

Star topology is more reliable, secure, and scalable, making it suitable for modern networks. Bus topology is economical but is mainly suitable for small networks due to its limited performance and reliability.

Question 2: Compare Mesh Topology and Star Topology

Introduction:

Mesh and Star topologies are widely used in computer networks. Mesh topology provides multiple communication paths, while Star topology uses a central switch for communication.

Comparison Table:

Parameter	Mesh Topology	Star Topology
Structure	Every device connects to multiple devices	All devices connect to one central switch
Fault Tolerance	Very High	Moderate
Reliability	Excellent	Good
Cost	Very High	Moderate
Installation	Complex	Easy
Performance	Excellent	High

Scalability	Difficult	Easy
Example	Banks, Military Networks	Offices, Colleges

Conclusion:

Mesh topology provides maximum reliability and fault tolerance, making it ideal for critical applications such as banking. Star topology is more affordable, easier to maintain, and widely used in organizations.

Question 3: Compare Bus Topology and Mesh Topology

Introduction:

Bus topology uses a single communication cable, whereas Mesh topology provides multiple dedicated communication paths. They differ significantly in complexity, cost, and reliability.

Comparison Table:

Parameter	Bus Topology	Mesh Topology
Structure	Single backbone cable	Every device connects to multiple devices
Cost	Low	Very High
Installation	Simple	Complex
Cabling	Less	More
Reliability	Low	Very High
Performance	Moderate	Excellent
Fault Tolerance	Low	Excellent
Example	Small Labs	Banking Networks

Conclusion:

Bus topology is suitable for small, low-budget networks, whereas Mesh topology is best for organizations requiring maximum reliability and continuous network availability.

Question 4: Compare Hybrid Topology and Star Topology

Introduction:

Hybrid topology combines two or more topologies to achieve flexibility and scalability, while Star topology uses a central switch to connect all devices.

Comparison Table:

Parameter	Hybrid Topology	Star Topology
Structure	Combination of multiple topologies	Central switch-based topology
Scalability	Very High	High
Flexibility	Excellent	Moderate

Cost	High	Moderate
Reliability	High	High
Installation	Complex	Easy
Maintenance	Moderate	Easy
Example	University Campuses	Offices

Conclusion:

Hybrid topology provides greater scalability and flexibility for large organizations, while Star topology remains the preferred choice for small and medium-sized networks because of its simplicity and lower cost.

Question 5: Compare the Maintenance of Star, Mesh, Bus, and Hybrid Topologies

Introduction:

Network maintenance refers to the ease of identifying faults, repairing connections, and expanding the network. Different topologies require different levels of maintenance.

Comparison Table:

Topology	Maintenance	Fault Detection	Reliability
Star	Easy	Easy	High
Mesh	Difficult	Moderate	Very High
Bus	Difficult	Difficult	Low
Hybrid	Moderate	Moderate	High

Explanation

- **Star Topology:** Easy to maintain because each device has an individual connection to the switch. Faults can be identified quickly.
- **Mesh Topology:** Difficult to maintain due to many cables and connections, but highly reliable because multiple paths exist.
- **Bus Topology:** Hard to troubleshoot since a backbone cable failure affects the entire network.
- **Hybrid Topology:** Maintenance is moderate because multiple topologies are combined and require skilled administration.

Conclusion:

Among all topologies, **Star topology** is the easiest to maintain, **Mesh topology** provides the highest reliability but is complex to manage, **Bus topology** is economical but difficult to troubleshoot, and **Hybrid topology** offers a balanced solution for large organizations requiring flexibility and scalability. These answers are written in a professional, assignment-ready format and cover the comparison points expected by your assessment rubric.